

Student Number: _____

NSW Edumacation Something Something

2021 YEARLY EXAM

Mathematics Extension 3

General

Instructions

- Reading time – 5 minutes
- Working time – 25 minutes
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total Marks:

25

Section I – 5 marks (pages 2–3)

- Attempt Questions 1–5
- Allow about 10 minutes for this section

Section II – 20 marks (pages 4–5)

- Attempt Questions 6–7
- Allow about 30 minutes for this section

Section III – 0 marks (pages 6–7)

- Attempt Questions 6–7
- Allow about 30 minutes for this section

Trigonometry

5 marks

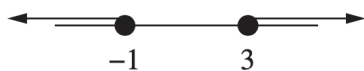
Attempt Questions 1–5

Allow about 10 minutes for this section

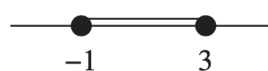
Use the multiple-choice answer sheet for Questions 1–5.

1 Which diagram best represents the solution set of $x^2 - 2x - 3 \geq 0$?

A.



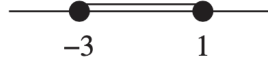
C.



B.



D.



2 Given $f(x) = 1 + \sqrt{x}$, what are the domain and range of $f^{-1}(x)$?

A. $x \geq 0, y \geq 0$

B. $x \geq 0, y \geq 1$

C. $x \geq 1, y \geq 0$

D. $x \geq 1, y \geq 1$

3 Which of the following is an anti-derivative of $\frac{1}{4x^2 + 1}$?

A. $2 \tan^{-1} \left(\frac{x}{2} \right) + c$

B. $\frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + c$

C. $2 \tan^{-1} (2x) + c$

D. $\frac{1}{2} \tan^{-1} (2x) + c$

- 4 Maria starts at the origin and walks along all of the vector $2\vec{i} + 3\vec{j}$, then walks along all of the vector $3\vec{i} - 2\vec{j}$ and finally along all of the vector $4\vec{i} - 3\vec{j}$.

How far from the origin is she?

- A. $\sqrt{77}$
- B. $\sqrt{85}$
- C. $2\sqrt{13} + \sqrt{5}$
- D. $\sqrt{5} + \sqrt{7} + \sqrt{13}$

- 5 The projection of the vector $\begin{pmatrix} 6 \\ 7 \end{pmatrix}$ onto the line $y = 2x$ is $\begin{pmatrix} 4 \\ 8 \end{pmatrix}$.

The point $(6, 7)$ is reflected in the line $y = 2x$ to a point A .

What is the position vector of the point A ?

- A. $\begin{pmatrix} 6 \\ 12 \end{pmatrix}$
- B. $\begin{pmatrix} 2 \\ 9 \end{pmatrix}$
- C. $\begin{pmatrix} -6 \\ 7 \end{pmatrix}$
- D. $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$

Section II

20 marks

Attempt Questions 6–7

Allow about 30 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 6–7, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (16 marks) Use the Question 6 Writing Booklet.

(a) Let $P(x) = x^3 + 3x^2 - 13x + 6$.

(i) Show that $P(2) = 0$ 1

(ii) Hence, factor the polynomial $P(x)$ as $A(x)B(x)$, where $B(x)$ is a quadratic polynomial. 2

(b) Use the principle of mathematical induction to show that for all integers $n \geq 1$, 4

$$1 \times 2 + 2 \times 5 + 3 \times 8 + \dots + n(3n - 1) = n^2(n + 1).$$

(c) The diagram shows the graph of $y = f(x)$. 3

Sketch the graph of $y = \frac{1}{f(x)}$.

Question 6 continues on page 5

Question 6 (continued)

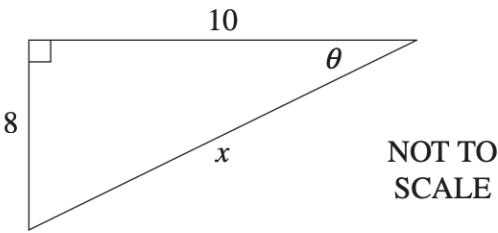
- (d) By expressing $\sqrt{3}\sin x + 3\cos x$ in the form $A\sin(x + \alpha)$, solve $\sqrt{3}\sin x + 3\cos x = \sqrt{3}$, for $0 \leq x \leq 2\pi$. 4
- (e) Solve $\frac{dy}{dx} = e^{2y}$, finding x as a function of y . 2

End of Question 6

Please turn over

Question 7 (4 marks)

Consider the triangle shown.



- (a) Find the value of θ , correct to the nearest degree. 2

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- (b) Find the value of x , correct to one decimal place. 2

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End of paper

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